

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, Golang, C++, Java, SQL

Tools/Frameworks: TensorFlow, PyTorch, FastAPI, Flask, Docker, Kubernetes, AWS SageMaker, OpenAI API, TensorRT, Airflow, PySpark, Hadoop, Hive, Sqoop, NiFi

Competence: Machine Learning, MLOps, LLMs, Deep Learning, AI Model Training, RAG, Data Engineering, Signal Processing, Data Privacy and Security, Feature Engineering

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed and evaluated deep learning models for metrological applications, enhancing model performance and training efficiency.
- Developed novel architectures and physics-informed hybrid loss functions to improve model accuracy and stability.
- Collaborated with PhD researchers to co-author a publication presented at the FLOMEKO conference.
- Built dashboards for data visualization and model performance monitoring.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Achieved 94% accuracy in vitiligo segmentation and improved inference speed by 5x using AI algorithms, deployed via Docker and FastAPI.
- Developed a real-time tennis foul detection system with TensorRT, reducing compute costs and time by 6x.
- Managed MLOps pipelines on AWS SageMaker, addressing YOLO model data drift, and maintained Kubernetes infrastructure.
- Conducted educational sessions on machine learning and big data at universities, mentoring students and professionals.
- Built POCs using YOLO, few-shot learning, and CNNs for client demonstrations.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Developed a full-stack website for AI-based services using Flask, HTML, and CSS.
- Built an AI chatbot application leveraging OpenAI API and LLM frameworks to enhance user interaction.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Designed and implemented web scrapers and ETL pipelines for structured and unstructured data.
- Automated data ingestion and cleaning workflows to streamline client analytics processes.

Projects

CAFA 6: Protein Function Prediction 2022

Deep Learning Competition

- Developed a weighted-probability ensemble combining DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction.
- Enhanced model accuracy by 4% through custom feature engineering across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting 2022

Thesis Project

- Built a real-time Bitcoin price forecasting tool using LSTM models, achieving a 15% improvement in prediction accuracy.

Saaz: AI Self-Playing Chitralli Sitar 2023

Signal Processing, Music Generation

- Collaborated to create the world's first AI-driven self-playing Chitralli Sitar using signal processing and music generation LLMs.
- Project featured on Coke Studio in a music video by Hassan Raheem, showcasing innovative AI applications in music.

Hugging Face Open-Source

Contributor

- Co-authored the Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub.

Education

Friedrich Alexander Universität Present

Master of Science in Artificial Intelligence *Erlangen, DE*

- Specialized in Machine Learning, MLOps, and AI-driven applications with a focus on LLMs and scalable solutions.

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science *Karachi, PK*

- Dean's List of Honor for academic excellence.
- Completed coursework in Data Science, Artificial Intelligence, and Software Engineering.

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python